

Step 1: Translate Strong Entities + Unnest Composite Attributes

```
CREATE TABLE Manager (  
    ssn INT PRIMARY KEY,  
    name INT,  
    email INT  
);
```

```
CREATE TABLE Driver (  
    name INT PRIMARY KEY  
);
```

```
CREATE TABLE Client (  
    name INT PRIMARY KEY,  
    email INT  
);
```

```
CREATE TABLE Car (  
    brand INT,  
    carid INT,  
    PRIMARY KEY (brand, carid)  
);
```

```
CREATE TABLE Rent (  
    rentid INT PRIMARY KEY,  
    date INT  
);
```

```
CREATE TABLE Address (  
    road_name INT,  
    number INT,  
    city INT,  
    PRIMARY KEY (road_name, number, city)  
);
```

```
CREATE TABLE CreditCard (  
    card_number INT PRIMARY KEY  
);
```

Step 2: Translate Weak Entities

Review (weak on Driver, ReviewID is the discriminator)

```
CREATE TABLE Review (  
    drivename INT,  
    reviewid INT,  
    rating INT,  
    message INT,  
    PRIMARY KEY (drivename, reviewid),  
    FOREIGN KEY (drivename) REFERENCES Driver(name)  
);
```

Model (weak on Car)

```
CREATE TABLE Model (  
    brand INT,  
    carid INT,  
    modelid INT,  
    color INT,  
    construction_year INT,  
    transmission_type INT,  
    PRIMARY KEY (brand, carid, modelid),  
    FOREIGN KEY (brand, carid) REFERENCES Car(brand, carid)  
);
```

Step 3: Translate Multi-valued Attributes

There don't appear to be any multi-valued attributes explicitly shown in the ER diagram, so **this step is skipped**.

Step 4: Translate Relationships

We will follow cardinalities and translation rules from your notes.

Client–CreditCard (1:1)

```
ALTER TABLE CreditCard
ADD client_name INT UNIQUE,
ADD FOREIGN KEY (client_name) REFERENCES Client(name);
```

Client–Address (0:1)

```
CREATE TABLE ClientAddress (
    client_name INT,
    road_name INT,
    number INT,
    city INT,
    PRIMARY KEY (client_name),
    FOREIGN KEY (client_name) REFERENCES Client(name),
    FOREIGN KEY (road_name, number, city) REFERENCES
Address(road_name, number, city)
);
```

Driver–Address (0:1)

```
CREATE TABLE DriverAddress (
    driver_name INT,
    road_name INT,
    number INT,
    city INT,
    PRIMARY KEY (driver_name),
    FOREIGN KEY (driver_name) REFERENCES Driver(name),
    FOREIGN KEY (road_name, number, city) REFERENCES
Address(road_name, number, city)
);
```

Rent–Client (1:1)

```
ALTER TABLE Rent
ADD client_name INT UNIQUE,
ADD FOREIGN KEY (client_name) REFERENCES Client(name);
```

Rent–Driver (1:1)

```
ALTER TABLE Rent
```

```
ADD driver_name INT UNIQUE,  
ADD FOREIGN KEY (driver_name) REFERENCES Driver(name);
```

Rent–CreditCard (HasPayment) (1:1)

```
ALTER TABLE Rent  
ADD card_number INT UNIQUE,  
ADD FOREIGN KEY (card_number) REFERENCES CreditCard(card_number);
```

Rent–Address (UsedIn) (0:1)

```
ALTER TABLE Rent  
ADD road_name INT,  
ADD number INT,  
ADD city INT,  
ADD FOREIGN KEY (road_name, number, city) REFERENCES  
Address(road_name, number, city);
```

Rent–Model (Has) (1:1)

```
ALTER TABLE Rent  
ADD brand INT,  
ADD carid INT,  
ADD modelid INT,  
ADD FOREIGN KEY (brand, carid, modelid) REFERENCES Model(brand, carid,  
modelid);
```

Car–Model (Has) (1:1) — Already handled via Model's FK.

Driver–Model (CanDrive) (0:N)

```
CREATE TABLE CanDrive (  
    driver_name INT,  
    brand INT,  
    carid INT,  
    modelid INT,  
    PRIMARY KEY (driver_name, brand, carid, modelid),  
    FOREIGN KEY (driver_name) REFERENCES Driver(name),  
    FOREIGN KEY (brand, carid, modelid) REFERENCES Model(brand, carid,  
modelid)  
);
```

Client–Review (Writes) (0:N)

```
ALTER TABLE Review
ADD client_name INT,
ADD FOREIGN KEY (client_name) REFERENCES Client(name);
```

Driver–Review (Receives) (1:1) — Already handled as part of Review's PK.

Manager–Car (Manages?) – Not shown in ER, but if relationship exists:

```
CREATE TABLE Manages (
    manager_ssn INT,
    brand INT,
    carid INT,
    PRIMARY KEY (manager_ssn, brand, carid),
    FOREIGN KEY (manager_ssn) REFERENCES Manager(ssn),
    FOREIGN KEY (brand, carid) REFERENCES Car(brand, carid)
);
```